1 An aluminium ring is placed on a coil with the rod of a metal retort stand passing through their centres, as shown in Fig. 1.1.

For Examiner's Use

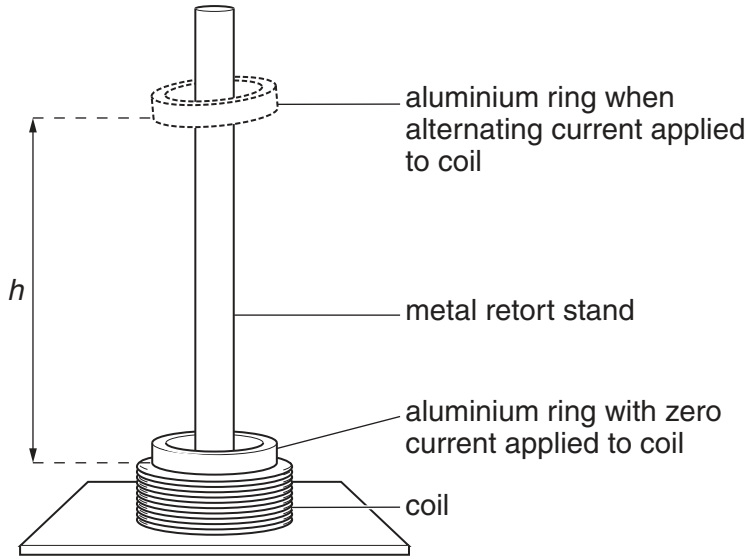

Fig. 1.1

When an alternating current of frequency f is applied to the coil, the ring rises until it is in equilibrium at a height h above the coil.

It is suggested that the relationship between h and f is

$$h = kf^n$$

where k and n are constants.

Design a laboratory experiment to test the relationship between h and f and determine values for k and n. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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